

## DYNAMIC FINITE ELEMENT ANALYSIS OF STRUCTURES

Mechanical Engineering 2024-2025 Master semester 2

Specialization/orientation: Mechanics of solids and structures

CFU: 3

Language: English

Semester: Spring

Lectures: 2 hours per week x 14 weeks

Exercises: 1 hour per week x 14 weeks

Exam: summer session (written)

**Summary.** The course covers linear 3D elastodynamics analysis of structures using the finite element method. Students will learn to construct the elemental stiffness, mass, damping, Coriolis, and angular and centrifugal acceleration matrices for rotating or non-rotating, conservative or dissipative systems. Beam elements and thin to moderately thick shell finite elements are then developed. Potential numerical issues (such as shear locking, rigid-body modes, etc.) are discussed, and solutions to overcome these difficulties are presented.

In the second part, the course focuses on solving the free response (modal analysis) of conservative structures with a large number of degrees of freedom using the subspace method. Finally, it addresses the solution of the forced response (time-domain analysis) of dissipative structures with many degrees of freedom using Newmark methods.

**Keywords.** Finite element, Beam and shell elements, Structural dynamics, Subspace method, Newmark methods

### Content.

- Strong and weak forms of the governing equations in solid mechanics
- Extension of the finite element method to 3D dynamic structural problems
- Development of solid and shell finite elements
- Strain matrix, local and global coordinate systems, assembly of local elements
- Development of modal parameter extraction methods
- Numerical analysis of the forced regime of structures
- Examples and case studies in MATLAB

### Learning Prerequisites.

- Finite element method
- Finite element modelling and simulation
- Structural mechanics
- Solid mechanics
- Mechanical vibrations

**Concepts importants à maîtriser.**

- Apply the concepts of structural dynamics to predict the vibratory behaviour, free or forced, conservative or damped, of continuous systems or systems with multiple degrees of freedom. (S3)
- Derive a finite element formulation of a physical problem from its strong-form differential equations (using the principle of virtual work or a variational approach). (S9)
- Implement the theoretical concepts of the finite element method to carry out a complete study of a real-world problem based on a project specification, and provide a critical analysis of the results obtained. (S10)

**Learning outcomes.** By the end of the course, the student must be able to:

- Mathematically formulate a structural dynamics problem corresponding to a real-world application; select and implement an appropriate modeling method and solution algorithm, whether for transient, harmonic, or modal dynamics, in either the physical or modal domain. (S15)
- Scientifically explain and implement a numerical method for multiphysics modeling and simulation appropriate to a complex physical problem. Abstract and derive a model of the dominant physical behaviors, along with the initial and boundary conditions. (S16)

**Transversal skills.**

- Self-assess one's level of acquired competence and plan future learning objectives
- Use both general and domain specific IT resources and tools.
- Write a scientific or technical report
- Deliver an oral presentation
- Plan actions and carry them out effectively to make optimal use of time and available resources

**Teaching methods.** Ex cathedra lectures, exercises in the classroom and computer lab sessions.

**Expected student activities.**

- Class participation
- Solving exercises and problems
- Completion of several mini-projects
- Writing scientific reports

**Assessment methods.** 50% mini-projects during the semester and 50% final oral exam

**Ressources.**

- Thomas Gmür, Méthode des éléments finis en mécanique des structures, Presses polytechniques et universitaires romandes (PPUR), Lausanne, 2018, ISBN 978-2-88915-158-5.

- Thomas Gmür, Dynamique des structures – Analyse modale numérique, Presses polytechniques et universitaires romandes (PPUR), Lausanne, 2014 (2ème édition), ISBN 978-2-88074-813-5
- Thomas J. R. Hughes, The Finite Element Method – Linear Static and Dynamic Finite Element Analysis, Prentice-Hall, Englewood Cliffs, NJ, 1987 (Dover Publications, 2000), ISBN 978-0-48641-181-1